LEAKLISTEN · TRACEABILITY MATRIX

# LeakListen traceability matrix

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## DOCUMENT CONTROL

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| Document ID | LKL-RTM-001 | Version | 0.4 |
| Status | Draft | Date | 2026-11-01 |
| Author | Amish Chadha | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept | | |
| Repository | github.com/BoujeeEnjinia1701/leaklisten | | |
| Source | docs/03b-traceability.md @ a81e84a | | |

## REVISION HISTORY

| Version | Date | Author | Change |
|---|---|---|---|
| 0.4 | 2026-11-01 | Amish Chadha | Generated by .kit/trace.py |
| 0.3 | 2026-11-01 | Amish Chadha | Generated by .kit/trace.py |
| 0.2 | 2026-11-01 | Amish Chadha | Generated by .kit/trace.py |
| 0.1 | 2026-11-01 | Amish Chadha | Generated by .kit/trace.py |

# LeakListen traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 16 user requirements, 20 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

| User requirement | Product requirement | Title | Evidence | Verification | Status |
|---|---|---|---|---|---|
| UR-001 | PR-001 | Hears a leak on metallic mains | LKL-CAL-001, LKL-STM-001 | Attenuation and detection estimate (LKL-CAL-001 section G); later a partner site with a controlled leak | At risk:** about 185 m on ductile iron at the central estimate (signal equal to noise), about 140 m if the leak must lift the level by the 6 dB of the flag rule (independent review, LKL-STM-001 S2); 93 to 370 m over plausible pipe losses |
| UR-002 | PR-002 | Hears a leak on plastic mains (hydrophone variant) | LKL-CAL-001, LKL-OPT-001 | As PR-001, for the hydrophone variant | Open: the hydrophone variant is not sized in LKL-CAL-001. First-order screen only (LKL-STM-001 SB1): 54 m on PVC at an assumed 0.1 Pa/rtHz background, none at 1 Pa/rtHz. For the record the contact sensor hears about 4 m on PVC and 2 m on PE |
| UR-001 | PR-003 | Sensor bandwidth | LKL-CAL-001, LKL-BLD-001 | Resonance estimate (LKL-CAL-001 section F); later a shaker comparison | At risk:** with the 42 mm magnet and the 8 mm puck base the mount resonates at about 676 to 1,350 Hz, flat only to 370 to 730 Hz |
| UR-001 | PR-004 | Sensor self-noise | LKL-CAL-001, LKL-BLD-001 | Noise calculation (LKL-CAL-001 section F) | At risk:** 1.03 µg/√Hz worst case with the 53 g seismic mass |
| UR-005 | PR-005 | Night listening | LKL-CAL-001 | Clock drift estimate (LKL-CAL-001 section H) | Met on paper with a weekly network time correction (20 s worst); 87 s a month free running |
| UR-007 | PR-006 | Battery life | LKL-CAL-001, LKL-BLD-001 | Power budget (LKL-CAL-001 section C) | Met on paper: 10.1 years or more on energy; taken as 10 years |
| UR-005 | PR-007 | Radio link | LKL-CAL-001, LKL-BLD-001 | Link budget (LKL-CAL-001 section D); cover loss measurement and field check later (TRL 4, on hold) | At risk:** +0.9 dB at 0.5 km under a 20 dB iron cover (EU868 SF12), -1.2 dB for US915 SF9, -9.1 dB for a tight 30 dB cover; cover loss unmeasured |
| UR-009 | PR-008 | Data and privacy | LKL-CAL-001 | Design review | Met by design |
| UR-003, UR-012 | PR-009 | Attachment | LKL-CAL-001, LKL-BLD-001 | Hold estimate (LKL-CAL-001 section I); a pull test later | Met on paper with the 42 mm magnet (LKL-DDR-002): 125 N on a 0.5 mm coating, 187 N on 0.3 mm. Non-ferrous fittings, outside this target, need an adapter |
| | PR-010 | | | | |

| User requirement | Product requirement | Title | Evidence | Verification | Status |
|---|---|---|---|---|---|
| UR-003, UR-004 | | Install from the surface | LKL-CAL-001, LKL-BLD-001, LKL-STM-001 | Task estimate (LKL-CAL-001 section K); a walk-through with the partner utility | Not verifiable at TRL 3; the 11 min estimate (neck bar assembled beforehand) is 1 min over the target, so it is at risk on the estimate; cable reaches caps 2.18 m down |
| UR-006 | PR-011 | Survive the chamber | LKL-CAL-001, LKL-BLD-001, BOM | Design review (LKL-CAL-001 section J) | Met by design; the logger floats on its lanyard when flooded |
| UR-014 | PR-012 | Size and mass | LKL-CAL-001, LKL-BLD-001, LKL-STM-001 | Parametric model (LKL-CAL-001 section K); weighing at TRL 4 | Met on paper: 68 x 283 mm overall; logger, cable and sensor 0.96 kg (0.04 kg margin); the neck bar is 0.31 kg. The 70 g antenna rides on the bar and is counted with it; with the antenna the carried set is 1.03 kg (independent review 2026-11-01): the exclusion is Proposed, awaiting Amish to confirm |
| UR-013 | PR-013 | Leak flag quality | LKL-CAL-001 | Field data from a partner network | Not verifiable at TRL 3 |
| UR-010, UR-015 | PR-014 | Open and interoperable | LKL-CAL-001, LKL-BLD-001 | Design review (LKL-CAL-001 sections A and B) | Met by design |
| UR-008 | PR-015 | Cost | LKL-CAL-001 | Priced BOM (LKL-CAL-001 section M) | USD 142, USD 12 over the value-engineering target |
| UR-003, UR-011 | PR-016 | No confined space entry | LKL-CAL-001 | Design review; build plan steps (LKL-BLD-001) | Met by design (surface placement by pole, PR-010) |
| UR-011 | PR-017 | Safe cell | LKL-CAL-001 | Design review; main board design at TRL 4 | Met by design on paper; the board is not designed at TRL 3 |
| UR-011, UR-016 | PR-018 | Labelled | LKL-CAL-001 | Inspection of BOM line 11 and the build plan | Met by design (label in BOM line 11) |
| UR-012 | PR-019 | Nothing touches the water | LKL-CAL-001 | Design review of the model (no part passes through a pipe wall) | Met by design; applies to option A only (option B's hydrophone is in the water, LKL-REG-001 G4) |
| UR-016 | PR-020 | Legal transmit power | LKL-CAL-001, | Design review (LKL-CAL-001 D4); | Met by design in EU868 at +14 dBm; the TRL 2 figure of +20 dBm |

| User requirement | Product requirement | Title | Evidence | Verification | Status |
|---|---|---|---|---|---|
| | | | LKL-STM-001 | regulatory check at TRL 4 (LKL-REG-001) | conducted is not permitted there unless the antenna gain and lead loss bring it under 14 dBm ERP (independent review): the free-field antenna gain is to assess |

## Gaps

None.